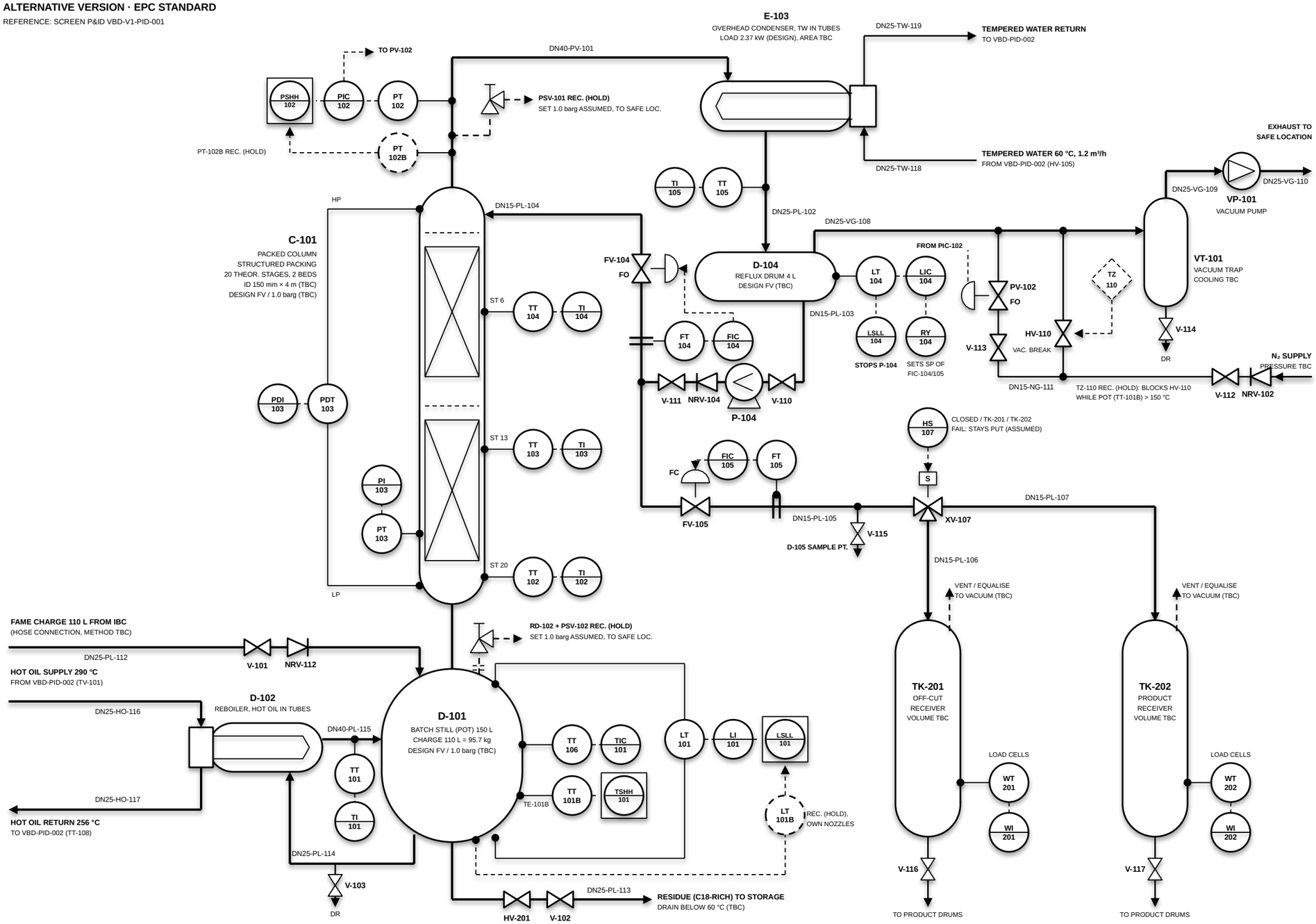
ALTERNATIVE VERSION · EPC STANDARD

REFERENCE: SCREEN P&ID VBD-V1-PID-001



LEGEND

- HAND VALVE (BLOCK / DRAIN), V-1nn
- CHECK VALVE, FLOW →, NRV-1nn
- GLOBE (BYPASS) VALVE
- CONTROL VALVE, DIAPHRAGM; FO / FC = FAIL OPEN / CLOSED
- ON/OFF VALVE, SOLENOID-PNEUMATIC (S)
- THREE-WAY DIVERTER VALVE
- RELIEF VALVE (DASHED = RECOMMENDED, HOLD)
- ORIFICE FLOW ELEMENT
- CENTRIFUGAL PUMP (MOTOR NOT SHOWN)
- FIELD INSTRUMENT (TRANSMITTER / ELEMENT)
- DCS FUNCTION (SHARED DISPLAY / CONTROL)
- SIS-100 FUNCTION (TRIP)
- RECOMMENDED INSTRUMENT (DASHED, HOLD)
- INTERLOCK LOGIC (DIAMOND)
- INSTRUMENT SIGNAL (ELECTRIC / SOFTWARE)
- PROCESS LINE, FLOW DIRECTION
- UTILITY / DRAIN / VENT LINE
- CONNECTION (TEE)

LINE NUMBER DNxx-SVC-1nn (SIZES PRELIMINARY, TBC)

PV PROCESS VAPOUR · PL PROCESS LIQUID · VG VACUUM GAS
NG NITROGEN · HO HOT OIL · TW TEMPERED WATER · CW COOLING WATER

EQUIPMENT LIST (BOTH SHEETS)

TAG	SERVICE	DATA (DESIGN CASE)
C-101	PACKED COLUMN	20 STAGES, ID 150 mm TBC
D-101	BATCH STILL (POT)	150 L, CHARGE 110 L
D-102	REBOILER, HOT OIL	DUTY 3.0 kW
E-103	OVERHEAD CONDENSER	2.37 kW, TW 60 °C
D-104	REFLUX DRUM	4 L; 50 % = 1.74 kg
P-104	REFLUX / DISTILLATE PUMP	DRAW 0–70 kg/h
VT-101	VACUUM TRAP	SIZE / COOLING TBC
VP-101	VACUUM PUMP	TOP 160 mbar a
TK-201	OFF-CUT RECEIVER	11.1 kg (CAMPAIGN A)
TK-202	PRODUCT RECEIVER	34.7 kg (CAMPAIGN A)
HE-101	HOT-OIL HEATER (PID-002)	12 kW ELECTRIC
P-101	HOT-OIL PUMP (PID-002)	125 kg/h AT DESIGN
E-104	HOT-OIL COOLER (PID-002)	COOL-DOWN ONLY
P-108	TEMP.-WATER PUMP (PID-002)	1.2 m³/h, 60 °C
TK-102	HOT-OIL EXP. TANK	RECOMMENDED, HOLD

GENERAL NOTES

1. ALTERNATIVE VERSION TO EPC STANDARD. THE SCREEN P&ID VBD-V1-PID-001 (P&ID TAB, TOP) STAYS THE REFERENCE FOR THE DCS, THE DECKS AND THE EXERCISES.
2. DESIGN CASE (OTS MODEL, NOISE OFF, START OF CAMPAIGN AT TOTAL REFLUX): TOP 160 mbar a (PIC-102), POT 170.9 mbar a, COLUMN ΔP 10.9 mbar, FLOODING 61 %; POT DUTY 3.0 kW (JIC-101); HOT OIL 290 °C (TIC-107); TEMPERED WATER 60 °C, 1.2 m³/h; CHARGE 110 L = 95.7 kg FAME (89 % LEVEL HOT, 133 L AT 255 °C).
3. TOTAL REFLUX: TI-104 192.4, TI-103 193.3, TI-102 239, POT 255.9 °C; OVERHEAD 31.7 kg/h (MW 214); E-103 LOAD 2.37 kW; REFLUX 31.8 kg/h.
4. CAMPAIGN A: OFF-CUT R 15 → TK-201, PRODUCT CUT R 10 → TK-202 (C16:0 ≥ 99.0 wt %). SWITCH 307 min, END 907 min; TK-201 11.1 kg, TK-202 34.7 kg, RECOVERY 80.7 %; RESIDUE 45.5 kg STAYS IN D-101. MASS BALANCE CLOSES TO 0.002 % (eng_check.py).
5. VAPOUR-PRESSURE ANCHORS, ΔHv, PACKING CAPACITY, CONDENSER UA, POT HEAT LOSS ARE ASSUMED IN THE MODEL (TBC).
6. LINE SIZES PRELIMINARY (TBC). OVERHEAD DN40: 7.6 m/s, 0.12 mbar/m (DN25 WOULD GIVE 17.9 m/s, 1.1 mbar/m). LIQUID LINES AT MINIMUM SIZE.
7. ALL VESSELS SEE VACUUM: DESIGN FV / 1.0 barg ASSUMED (HOLD); MATERIALS TBC. ESTER LINES TRACED / INSULATED ABOVE 45 °C (ESTERS FREEZE 30–38 °C), TBC.
8. PRESSURE AND LEVEL INSTRUMENTS C/W ROOT AND TEST VALVES (TYP., NOT SHOWN). PUMP MOTORS AND CASING DRAINS NOT SHOWN.
9. NEW TAGS ON THIS SET: V-101…V-122, NRV-101/102/104/108/112, FT-101, TT-101, LINE NUMBERS. ALL OTHER TAGS ARE THE OTS TAGS, UNCHANGED.
10. CHARGE METHOD, RECEIVER VENT / EQUALISING LINE, VT-101 COOLING AND DRAINING: TBC.

INSTRUMENT AND SAFETY NOTES

11. FIELD TRANSMITTERS (CIRCLE) ARE DRAWN SEPARATE FROM DCS FUNCTIONS (BAR) AND SIS-100 FUNCTIONS (SQUARE). AS IN THE MODEL, PSHH-102 SHARES PT-102 AND LSLL-101 SHARES LT-101 WITH THE DCS; TSHH-101 HAS ITS OWN ELEMENT TE-101B.
12. SIS-100 TRIPS (DELAY): TSHH-101 > 290 °C (5 s) · PSHH-102 > 300 mbar (5 s, RUN) · FSLL-108 < 0.36 m³/h (10 s, RUN) · LSLL-101 < 20 % (5 s, RUN) · TSHH-107 > 330 °C (5 s) · HS-100. ACTIONS, PERMISSIVES AND RESET: INTERLOCK SCHEDULE ON VBD-PID-002.
13. LSLL-104 < 5 % FOR 5 s STOPS P-104 (PUMP PROTECTION IN THE BPCS, NOT A SIF).
14. FAIL POSITIONS: PV-102 FO (N_2 BLEED OPEN; TOP SETTLES ≈ 300 mbar = PSHH-102), FV-104 FO (REFLUX KEPT), FV-105 FC (NO DRAW-OFF), XV-101 / TV-101 FC (VBD-PID-002). XV-107 STAYS PUT AND HV-105 FO: ASSUMED (TBC).
15. PIC-102 → PV-102. LIC-104 → RY-104 → SET POINTS OF FIC-104 (R/(R+1)) AND FIC-105 (1/(R+1)). HS-107 → XV-107: CLOSED / TK-201 / TK-202; A TRIP SETS CLOSED.
16. HV-110 N_2 VACUUM BREAK ONLY WITH THE POT BELOW 150 °C (PROCEDURE TODAY).
17. DASHED ITEMS ARE RECOMMENDED BY THE HAZOP AND SAFETY-LAYERS TABS AND DO NOT EXIST: PSV-101, RD-102 + PSV-102, PT-102B, LT-101B, TZ-110 (HOLD). RELIEF SET 1.0 barg ASSUMED; SIZE TO API 520/521 ONCE MAWP AND N_2 REGULATOR PRESSURE ARE KNOWN (HOLD).
18. RO-102 (N_2), TSV-103 (E-103 WATER SIDE), AT-102 (O_2) AND FA-101 ARE FURTHER RECOMMENDATIONS IN THE SAFETY-LAYERS TAB, NOT DRAWN.

REV	DESCRIPTION	DATE
—	VBD-V1-PID-001 A (SCREEN): CURRENT, DCS/DECKS	—
A	ALTERNATIVE VERSION, EPC STANDARD	12-OCT-26

CLIENT: DEPARTMENT OF CHEMICAL ENGINEERING, FACULTY OF ENGINEERING, CHULALONGKORN UNIVERSITY
VACUUM BATCH DISTILLATION VER 1 · OTS

P&ID — VACUUM BATCH DISTILLATION C-101 · ALTERNATIVE VERSION

DRAWN: Apinan Soottitantawat	CHK'D: --	APP'D: --	DATE: 12-OCT-26
DWG NO. VBD-PID-001	REV A	SCALE N.T.S.	SHEET 1 OF 2